ENGI 9839: Course Project
Software Verification and Validation

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The course project for ENGI 9839 is designed to strengthen students' understanding of Software Verification and testing through independent inquiry, critical analysis, and a small implementation component. Students will explore an approved topic related to software testing, formal methods, software verification, or software quality assurance, and will demonstrate their understanding through a written report, practical work, and a short presentation.

1. Project Structure

The course project may be completed individually or in groups of up to 2 students. Each project must address a topic approved by the instructor in advance. The topic should be related to the themes of the course but should not duplicate material already covered in class. Instead, it should either extend a topic discussed in the course or investigate an additional topic within software verification and testing.

2. Project Report [12%]

2.1 Report Requirements

The project report should be at least 12 pages in length, using a maximum of 12-point font and, where appropriate, double spacing. All sources, including research papers, books, and online references, must be cited using IEEE citation format. The report should present a coherent discussion of the selected topic, its relevance to software verification and testing, and the methods or tools used in the project.

2.2 Topic Expectations

The selected topic must be:

- Relevant to software verification and testing.
- Either a topic not covered in the course or a meaningful extension of a course topic.
- Approved by the instructor before the student or group begins the project work.

2.3 Implementation Component

The project must include a practical component, which may involve one or more of the following:

- A small software prototype.

- A formal specification model.
- A testing or analysis script.
- The application of a verification tool to a software system or case study.

The submission must include:

- Source code, model files, or other relevant implementation artifacts.
- Clear instructions on how to run the project.
- Sample output or results.
- A brief explanation of the implementation and its findings.

3. Presentation [8%]

3.1 Presentation Format

Each student or group will deliver a presentation of approximately 6 minutes, followed by 2 minutes for questions and answers. The presentation should clearly summarize the project topic, motivation, approach, results, and conclusions.

3.2 Slide Requirements

Students must submit a copy of their slides. The slides should reflect academic integrity standards and must include citations where appropriate. A bibliography in IEEE format should be included in the slide deck.

3.3 Literature Summary

In addition to the presentation, students must submit a brief discussion of the literature used to prepare the presentation. This should include:

- A one- to three-paragraph summary of each source.
- A complete bibliography in IEEE format.

4. Important Dates

- Topic selection deadline: May 24
- Presentation period: Beginning July 20

Each student or group should email the instructor with their chosen topic by the stated deadline. Topics will be assigned on a first-approved basis, and only one student or group may work on a given topic.

5. Suggested Topics

Students may choose any approved topic that falls broadly within the themes of software verification and testing. Examples of suitable topics include:

- Software testing techniques and test design methods.
- Formal specification using TLA+.
- Model checking for concurrent or distributed systems.
- Static analysis techniques for defect detection.
- Mutation testing and test adequacy assessment.
- Symbolic execution and automated test generation.
- Runtime verification and monitoring.
- Verification of safety- or security-critical software.
- Verifying software behavior in concurrent or distributed systems.
- Comparing manual testing and automated verification approaches.
- Applying formal methods to real-world case studies.
- Testing and verification of AI-based software systems.

6. Project Scope

Students may select an existing software system, a small prototype, or a case study as the subject of their project, provided that the work applies one or more software testing or verification techniques to analyze, evaluate, or formally specify the system.